

How teaching quality dimensions influence student satisfaction in entrepreneurship and innovation education: evidence from Jordan

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Abstract

Purpose – This study examines the influence of perception of teaching quality dimensions, namely course objectives, course content, teaching methods, teaching environment, course management and evaluation methods, on student satisfaction for innovation and entrepreneurship education.

Design/methodology/approach – This study uses a quantitative survey design at a single Jordanian university. Data were gathered using an item-based questionnaire. The survey was distributed to the student population (2,500 students), with the goal of conducting a census. However, due to the voluntary nature of participation, the final sample of 486 respondents was chosen by self-selection. The data were analyzed with SPSS, using multiple regression and reliability analyses to investigate the relationship between variables.

Findings – Multiple regression analysis revealed a positive and significant effect of educational content, teaching methods, teaching management and evaluation methods on student satisfaction. However, learning objectives were found to have an insignificant effect on student satisfaction. Furthermore, findings demonstrate that teaching conditions negatively impact student satisfaction.

Originality/value – To the researchers' best knowledge, this study is the first to examine the impact of perception of teaching quality dimensions on student satisfaction in the Middle East, with a specific focus on Jordan. Additionally, this study presents course objectives and evaluation methods as crucial teaching quality dimensions that were not previously investigated in the entrepreneurship and innovation education context. In addition, the study enhances understanding of student satisfaction in Jordanian higher education by creating a satisfaction model that is appropriate for the local culture.

Keywords Teaching quality, Entrepreneurship education, Innovation education, Student satisfaction, Jordan

Paper type Research article



1. Introduction

The increase of innovation and entrepreneurship as critical drivers of economic growth, job opportunities, and societal advancement has been witnessed in the twenty-first century. Internationally, academic institutions are being forced to modify their curricula and teaching techniques to suit the developing needs of the global economy. Education on entrepreneurship and innovation has become a fundamental element for providing learners with the necessary

skills, expertise, and attitudes required to succeed in an ever-changing environment (Boldureanu *et al.*, 2020). In Jordan, entrepreneurship and innovation education has emerged as a priority focus area because it addresses the pressing issues of youth unemployment, economic diversification, and the creation of an innovation ecosystem. For instance, even with the many entrepreneurship courses offered by Jordanian universities, studies still show that students are mainly exposed to theoretical content and traditional teaching methods instead of applied, skill-based approaches to learning (Hanandeh *et al.*, 2021). Additionally, curricular content tailored to entrepreneurship and innovation has been shown to significantly build students' entrepreneurial knowledge and processes in the Jordanian context (Alakaleek, 2019). Therefore, it is important to examine the relationship between the quality of teaching dimensions and their impact on student satisfaction for entrepreneurship and innovation courses. Consequently, this study examined those relationships and student satisfaction in the context of a Jordanian university to inform how higher education programs might better align with national innovation and youth employment policies.

An important aspect of evaluating the effectiveness of entrepreneurship and innovation education programs at universities is understanding student satisfaction with the quality of these courses (Abualbasal and Badran, 2019; Gao *et al.*, 2021). Their satisfaction not only reflects how students feel about these courses and materials but is a key measure of how well these programs meet the aspirations and expectations of the next generation of business leaders, visionaries, and decision-makers (Kanwar and Sanjeeva, 2022). Students' satisfaction with their education in the field of innovation and entrepreneurship reflects how well educational institutions are ready for them to develop under the modern business landscape and contribute to economic and social progress (Boldureanu *et al.*, 2020). It is vital to highlight that entrepreneurial education does not just strive to educate students on how to become business managers; it also teaches them how to solve issues creatively once they arise (Abualbasal and Badran, 2019).

We investigate the factors that contribute to their satisfaction and how educational institutions can enhance exceptional innovation and entrepreneurship programs to better meet the desires and expectations of their students. Assessing satisfaction levels requires understanding students' views on the learning process. This paper addresses gaps in the literature by examining perceptions of quality dimensions and their impact on student satisfaction with innovation and entrepreneurship education. Focusing on the Middle East, particularly Jordan, we also offer recommendations for institutions and policymakers to ensure program alignment with student needs and a dynamic business environment. Consequently, the study constructed this research question (RQ):

RQ. What dimensions influence student satisfaction with the perception of teaching quality of innovation and entrepreneurship courses?

The significance of this investigation extends beyond academics since it guides educational policymakers, institutions, and stakeholders on how to improve their innovation and entrepreneurship education offerings (Gao *et al.*, 2021). The study encourages students to express their ideas and concerns and fosters active communication between teachers and students. It evaluates course quality by considering student perceptions of the course, teaching strategies, teacher support, and availability of resources. As innovation and entrepreneurship continue to shape the global economy, it is important to adapt to these changing requirements.

The key contribution of this research to the existing literature lies in providing educational institutions with practical recommendations on how to improve the quality of their curricula. In addition, the research will seek to study variables that have been addressed to a limited extent by previous literature in the context of innovation and entrepreneurship education, which contribute to student satisfaction and teaching quality (Huang *et al.*, 2022; Gao *et al.*, 2021), such as course objectives and evaluation methods. Studies examining how teaching quality elements are perceived and their impact on students' satisfaction with innovation and

entrepreneurship education remain limited, particularly in developing nations ([Al Matroushi et al., 2021](#); [Huang et al., 2022](#)). These realities and the need to gain further insights into the dimensions of the quality perception related to students' satisfaction with innovation and entrepreneurship education have motivated the present study, particularly in the Jordanian context and other developing countries ([Gao et al., 2021](#)). Furthermore, recent research in Jordan has revealed several recurrent difficulties that have a direct impact on teaching quality and student satisfaction. These include insufficient evaluation techniques, inconsistent teaching methods, limited technological infrastructure, and decreased student-teacher interaction—all of which were particularly noticeable during the COVID-19 pandemic ([Almahasees et al., 2021](#); [Haider and Al-Salman, 2020](#)). Studies have shown that, while online learning platforms provided continuity, low system quality and a lack of contact harmed students' learning outcomes and satisfaction ([Mohammed et al., 2024](#); [Al-Soub et al., 2021](#)). Even before the pandemic, research found that faculty support and educational service quality at public universities were insufficient to meet students' expectations ([Aljarrah and Khataybeh, 2022](#)).

2. Literature review

2.1 Perception of teaching quality

Multiple studies focus on perceptions of teaching quality around the world, but there are still areas that need further investigation. The subject of perceptions of teaching quality has been investigated from various viewpoints, with each researcher studying the topic according to their needs ([Gao et al., 2021](#); [Haughney et al., 2020](#)). Some studies found that students define teaching quality as clarity, involvement, and the mode of instruction ([Gao et al., 2021](#); [Bijlsma and Röhl, 2023](#)).

Perceptions of teaching quality are critical to assessing the performance of educational systems ([Bijlsma and Röhl, 2023](#)). It significantly influences college students' learning experiences and directly impacts their academic performance ([Darling-Hammond et al., 2019](#)). According to [Lizzio et al. \(2002\)](#), the perception of teaching quality is a complex construct influenced by various elements, including engagement, feedback, instructional clarity, and the alignment of course objectives and evaluations. [Trigwell and Prosser \(2004\)](#) highlight the importance of both teacher-centered and student-centered teaching techniques.

Other researchers have found that teaching materials, teaching techniques, and administration all have a substantial impact on student satisfaction ([Weerasinghe and Fernando, 2018](#); [Gao et al., 2021](#); [Gunn, 2018](#)). [Wisenthige et al. \(2025\)](#) stated that students at private universities rated academic staff quality, including their subject knowledge, teaching style, and student support, as the most important factor influencing perceived teaching quality. In Pakistan, [Ikram and Kenayathulla \(2023\)](#) highlighted the significance of classroom setup, technology preparedness, and instructional materials in influencing how well students perceive their teachers. In addition, [Hmoud et al. \(2024\)](#) investigated the use of gamification in Jordanian and Palestinian higher education. It has been demonstrated that implementing gamification techniques, such as leaderboards, badges, and points, improves students' academic performance and raises opinions about teachers' effectiveness. [Firdousi et al. \(2024\)](#) found that teaching quality has a positive impact on student satisfaction. A focus is placed on variables related to student satisfaction that have been investigated to a limited extent by researchers, in order to fill gaps in the literature ([Esteban and del Cerro, 2020](#); [Weerasinghe and Fernando, 2018](#)).

In the present study, perceptions of teaching quality are operationalized using the most widely used and cited dimensions in the literature, which serve as the foundational principles for understanding teaching quality within the field of educational research. These include course objectives, teaching content, teaching methods, teaching conditions, teaching management, and evaluation methods ([Darawong and Widayati, 2022](#); [Darling-Hammond, 2021](#)).

2.2 Student satisfaction

Over the past few decades, researchers have focused on student satisfaction as it significantly impacts educational institutions and student enrollment. Numerous researchers have described student satisfaction as a temporary attitude that arises from an evaluation of a student's learning experiences (Kanwar and Sanjeeva, 2022; Weerasinghe and Fernando, 2018). Therefore, the performance of higher education institutions that offer educational services and varied perceived degrees of experience quality may be considered factors influencing student satisfaction levels (Weerasinghe and Fernando, 2018; Weerasinghe *et al.*, 2017). Besides, student satisfaction is a multifaceted concept influenced by several variables, such as academic experiences, institutional support, and the degree of alignment of student expectations to actual learning outcomes, affecting student satisfaction (Elliott and Shin, 2002; Astin, 1993). Research indicates that course relevance, administrative responsiveness, and resource availability all have an impact on student satisfaction in addition to teaching quality (Douglas *et al.*, 2006).

In Finland, the overall satisfaction levels of students and staff are more significantly influenced by research and teaching facilities, which are basic activities of universities, than by supported facilities (Kärnä and Julin, 2015). In Saudi Arabia, Alyoussef and Omer (2023) found that self-directed learning and technology-enhanced learning in Saudi Arabia improve undergraduate students' satisfaction with online learning. In Sri Lankan private universities, student satisfaction is significantly influenced by various factors such as university facilities, academic staff, degree programs, institution reputation, fees, and overall service quality (Wisenthige *et al.*, 2025). In India, Thomas (2025) examined the relationship between institutional rankings and student satisfaction in Indian universities and concluded that higher rankings do not always translate into satisfied students.

In Jordan, a limited number of researchers investigated the topic of student satisfaction, as many areas are still unaddressed in the literature (Al-Haddad *et al.*, 2018). Hence, this study extends prior literature by providing empirical evidence regarding the perceived teaching quality factors influencing student satisfaction in Jordan with a particular focus on innovation and entrepreneurship courses.

2.3 Student satisfaction models in higher education institutes

In higher education, several customer satisfaction survey methods are utilized, such as SERVQUAL and Happy-Productive Theory (Weerasinghe *et al.*, 2017). It is challenging to measure student satisfaction levels using the business-focused service quality model since government universities operate in the non-profit service sector. Service providers' quality is only one of several elements that influence the satisfaction of students in a university context. Nevertheless, all models have been assessed, and because of cultural and contextual differences, they all have advantages, disadvantages, and limitations. This study will adapt the Input-Process-Output (IPO) Model, which has been investigated to a limited extent in previous literature, and Classroom Dynamics Theory, due to the limited research focusing on assessment fairness. The Input-Process-Output (IPO) model is a conceptual framework for understanding a variety of systems, including the educational system. The Inputs include the study objective and course material. The processes include teaching methods and teaching conditions. Finally, outputs include teaching management and student assessment and satisfaction (Lizzio *et al.*, 2002). Classroom Dynamics Theory is a novel approach that bridges the gap between assessment and instruction in the classroom setting. This theory focuses on educational fairness, interactional fairness, distributive fairness, egalitarian assessment, and equitable assessment (Rezai, 2022).

2.4 Student satisfaction model for jordanian higher education

Due to the hybrid admission system, which includes two primary pathways for admission, the Jordanian university system differs significantly. The first is through the Ministry of Higher

Education, offering low or subsidized tuition fees. The second option is direct university admission, which typically requires higher tuition fees than the official government pathway. To fill this gap, the authors developed the “Jordanian Student Satisfaction Model for Higher Education (JSSMHE)” (Figure 1) by integrating six dimensions from the literature and adapting the Input-Process-Output (IPO) model and Classroom Dynamics Theory which integrates students engagement with an emphasis on comprehending the factors influencing course effectiveness. It incorporates the perspectives of diverse students into the fairness assessment process within educational environments:

- (1) Quality of course objectives
- (2) Quality of course content
- (3) Quality of teaching methods
- (4) Quality of teaching conditions
- (5) Quality of teaching management
- (6) Quality of evaluation methods

3. Theoretical framework and hypotheses development

3.1 Research framework

The current study aims to examine the influence of perceptions of teaching quality dimensions on student satisfaction. Moreover, our research methodology assumes that integrating the perception of educational quality dimensions will enhance student satisfaction. In this study, the Input-Process-Output (IPO) model (Ilgen *et al.*, 2005; Mathieu *et al.*, 2008) will be utilized as a framework, with key aspects from Classroom Dynamics Theory (Fraser, 1998) added to examine the effects of instructional factors on student satisfaction. The IPO model contextualizes the research by considering course objectives, content taught, teaching, teaching conditions, teaching management as inputs, teaching techniques, and evaluation methods as processes, and student satisfaction as output. Classroom Dynamics Theory, highlighting how the physical, procedural, and organizational aspects of the classroom influence students’ perceptions and overall satisfaction, enhances this framework. In the context of teaching entrepreneurship and innovation, this combined approach provides a more nuanced understanding of how particular instructional factors and processes contribute to

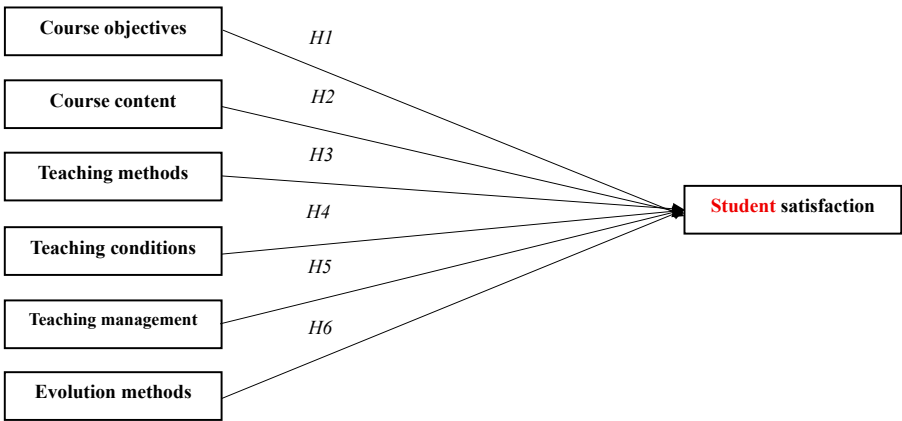


Figure 1. The research model

student satisfaction. Additionally, Classroom Dynamics Theory provides a classroom-specific perspective by focusing on the environment, including teaching practices, conditions, and assessment methods. Together, these theories and models form a complete theoretical foundation for investigating how teaching quality dimensions influence student satisfaction. [Figure 1](#) shows the suggested study model.

3.2 Quality of course objectives and student satisfaction

The term “course objectives” refers to how well students believe the intended learning outcomes are explained and how applicable they are to their academic and real-world learning requirements. Operationally, this construct shows how well students comprehend, value, and concur with the instructor’s or the syllabus’s stated course objectives ([Martin and Bolliger, 2022](#)). Course objectives are measurable goals that students aim to achieve by the end of the course ([Gao et al., 2021](#)). Many studies revealed that course objectives positively influence student satisfaction ([Darawong and Widayati, 2022](#); [Gray and DiLoreto, 2016](#)). Furthermore, other studies showed that when course objectives do not promote a student-centered learning strategy, student satisfaction could be negatively affected. In addition, in educational design, course objectives must be clear and effective. A well-defined objective offers an idea of the desired outcome, and SMART objectives provide a framework for goal setting in education, with specific criteria that make them easier to define, measure, and achieve ([Gray and DiLoreto, 2016](#)). Hence, we hypothesize the following:

- H1. The quality of course objectives will positively influence student satisfaction levels at the University of Jordan.

3.3 Quality of course content and student satisfaction

The term “teaching content” describes how students evaluate the practicality of the course’s instructional material, depth, clarity, and currency. It indicates how well the material supports students’ understanding of important ideas, how well it fits with the goals of the course, and how well it can be applied in professional or real-world settings. Students’ opinions about how thorough, current, and intellectually stimulating the course materials were served as an indicator for the construct ([Tharapos et al., 2023](#)). Previous research has demonstrated that the quality of teaching content positively affects student satisfaction in higher education ([Darawong and Widayati, 2022](#); [Ikram et al., 2025](#)). Also, the quality of teaching content should be understood and improved to enhance student satisfaction as well as their academic achievements ([Darawong and Widayati, 2022](#)). Understanding content types, selection criteria, delivery methods, and the use of advanced technology enables educators to design engaging learning experiences ([Tharapos et al., 2023](#)). Therefore, we formulate this hypothesis:

- H2. The quality of course content will positively influence student satisfaction levels at the University of Jordan.

3.4 Quality of teaching methods and student satisfaction

Teaching methods refer to the strategies and approaches used by teachers to present course content and enhance learning. This includes the use of interactive lectures, multimedia tools, group projects, case studies, and other educational methods designed to engage students, foster critical thinking, and improve comprehension. In this study, the construct is measured by students’ evaluations of how well teaching tactics assisted them in learning, encouraged involvement, and kept them interested in the course ([Alqasa and Afaneh, 2022](#)). Student satisfaction in higher education institutions is significantly affected by the quality of teaching techniques used ([Ikram et al., 2025](#)). Research shows that teaching methods influence student

satisfaction, with key variables including the depth of material delivery, the use of knowledge, and the integration of new technologies and resources (Pandita and Kiran, 2023). Similarly, innovative pedagogies like modeling and instructional demos were found to be positively related to student satisfaction with instructional quality (Akram et al., 2022). So, we propose the next hypothesis:

- H3. The quality of teaching methods will positively influence student satisfaction levels at the University of Jordan.

3.5 Quality of teaching conditions and student satisfaction

The term “teaching conditions” refers to how students evaluate the logistical, technological, and physical characteristics of the classroom setting, which can assist or impede learning. This includes the availability and sufficiency of facilities, furnishings, classroom comfort, noise levels, and the overall environment that promotes effective teaching. The study’s teaching conditions are operationalized based on students’ ratings of classroom facilities and environmental variables that influence focus and involvement (Ikram et al., 2025). The quality of the educational environment significantly impacts both student satisfaction and academic performance (Kassab et al., 2024). A positive academic atmosphere fosters students’ enjoyment of learning, leading to improved academic and non-academic outcomes (Gao et al., 2021). Moreover, the successful relationships between professors and students are critical components that influence student satisfaction (Gruber et al., 2012). Thus, high-quality teaching conditions are necessary for fostering a positive learning environment and enhancing student satisfaction. Accordingly, the study constructs the current hypothesis:

- H4. The quality of teaching conditions will positively influence student satisfaction levels at the University of Jordan.

3.6 Quality of teaching management and student satisfaction

In this study, “teaching management” refers to curricular instructional management, encompassing components such as credit hours, course selection, and scheduling. These curriculum-related components, managed by the institution, are beyond students’ control (Darling-Hammond et al., 2019; Gao et al., 2021). According to Xiao et al. (2017), one of the primary reasons for China’s poor innovation and entrepreneurship education development is how quality management is implemented. This type of instructional administration, which excludes student input, can negatively impact teaching quality. Darling-Hammond et al. (2019) and Gao et al. (2021) revealed that the successful use of classroom instructional management techniques is essential for raising the quality of instruction and learning. To improve teaching quality, promote teaching improvements, and develop talent training programs, course assessment is an essential component of course teaching management (Zhou et al., 2024). Course management serves as the core component of teaching management by providing feedback on the teaching process (Ikram et al., 2025; Xiao et al., 2017). Hence, the following hypothesis is proposed:

- H5. The quality of teaching management will positively influence student satisfaction levels at the University of Jordan.

3.7 Quality of evaluation/assessment methods and student satisfaction

Evaluation methods refer to the approaches and resources that teachers use to evaluate students’ performance, learning, and achievement of course objectives (Levy-Feldman, 2025). This category includes formative assessments (quizzes and assignments), summative

assessments (final exams and projects), and alternative assessments (portfolios and presentations). Students' perceptions of the fairness, diversity, clarity, and alignment of assessment procedures with course material and learning objectives are assessed to operationalize the evaluation methods utilized in this study (Wondim and Dessie, 2024). The quality of evaluation and assessment methods has a considerable influence on student satisfaction in higher education (Ikram and Kenayathulla, 2023; Kanwar and Sanjeeva, 2022). Lecturers' ability to deliver learning materials, the quality of classroom facilities, and effective assessment and feedback methods all significantly impact student satisfaction levels (Rigopoulos, 2022). Improved assessment procedures, such as clear marking criteria and constructive feedback, enhance student satisfaction and academic experience, highlighting the need for ongoing development in assessment practices (Mate and Weidenhofer, 2022; Rawlusyk, 2018). Therefore, the study proposes the following hypothesis:

- H6. The quality of evaluation/assessment methods will positively influence student satisfaction levels at the University of Jordan.

4. Method

4.1 Data collection and sample

The survey for this study was developed using Google Forms and distributed to students enrolled in the required Entrepreneurship and Innovation course at the University of Jordan during the summer term of the 2022/2023 academic year. The course was conducted in a blended learning format, combining in-person sessions with the Moodle platform. The study population comprised 2,500 students, and given its manageable size and accessibility through Microsoft Teams, the survey was distributed to all students, aiming for a census-based approach. However, due to the voluntary nature of participation, the final sample of 486 respondents was obtained through self-selection. This approach ensured broad coverage across the university, while allowing students to decide whether to participate, resulting in a diverse sample across academic years and specializations. Participants were assured of anonymity and confidentiality to minimize response bias. Participation was entirely voluntary, with no personally identifiable information collected. To further mitigate bias, the questionnaire items were phrased in a neutral manner to avoid leading questions or socially desirable responses. Respondents were explicitly informed that there were no right or wrong answers, encouraging honest and thoughtful participation. To maximize response rates, the questionnaire was distributed through multiple channels, including Microsoft Teams, Moodle, and WhatsApp groups, with the support of course instructors and supervisors. Since participation was voluntary, the final sample size was determined by students' willingness to participate rather than a predefined sampling strategy. The data collection process spanned approximately three weeks (August 1–20, 2023), and the final respondent sample included students from diverse academic backgrounds, years, and specialties. Table 1 presents the demographic characteristics of the study sample.

4.2 Research instrument

To achieve the study objectives, the authors developed a self-administered questionnaire to collect primary data. The constructs used in this study were adapted from earlier related studies published in English. The initial version of the survey was developed in English and later translated into Arabic by the authors. The authors confirmed that all adapted scales demonstrated acceptable validity and reliability in their original publications. Thus, the selected measurement scales were assumed to have content validity. To measure students' perception of teaching quality and its dimensions, two items were adapted for the course objectives from Martín Rodríguez *et al.* (2019). Seven items were adapted from Gao *et al.* (2021) to measure educational content. Four items were adapted from Gao *et al.* (2021) and

Table 1. Measurement items for the study constructs

Item code	Measurement statement
Course objectives	
Co1	The course of entrepreneurship and innovation has clear goals
Co2	The objectives of the entrepreneurship and innovation course have quality criteria
Course content	
CC1	The course of entrepreneurship and innovation offers rich theoretical content
CC2	The entrepreneurship and innovation course offers distinctive applied content
CC3	The course on entrepreneurship and innovation has advanced and modern educational content
CC4	The course on entrepreneurship and innovation has consistent and organized educational content
CC5	The methods of teaching entrepreneurship and innovation courses connect theory and practice
CC6	The material provides a rich vocabulary and terminology related to entrepreneurship and innovation
CC7	There are practical application examples presented in each part of the material
Teaching methods	
TM1	The teaching of the course is based on highly qualified and skilled teaching staff
TM2	The teachers are interested in training students in innovative and entrepreneurial processes
TM3	The methods of teaching the material are in harmony with its goals and the nature of its content
TM4	The methods of teaching consider the individual differences between students
Teaching conditions	
TC1	The course of entrepreneurship and innovation has the availability of basic teaching tools such as photos, diagrams, videos, and presentations
TC2	The number of lectures allocated for the material is proportional to its nature and goals
TC3	There are constant communication channels between students and teachers
TC4	The e-learning platform at the University fits the nature and purposes of the material
TC5	The blended learning style used in the material is appropriate for the material's nature, substance, and teaching goals
Teaching management	
TMA1	Appropriateness of the curriculum schedule of the course
TMA2	Suitability of the student's chosen method of registration in the course
TMA3	Appropriateness of the study plan for the course
Evaluation methods	
EM1	The exams have transparency
EM2	The exam for the course has reliability
EM3	The content of the exam fits into the content of the material
EM4	The exam for the subject is characterized by diversity in terms of the form of questions and the results that measure them
EM5	The assessment methods for the material are convenient (assignments/projects/final exams)
Student satisfaction	
SS1	I am satisfied with the quality of this course
SS2	I advise university students to register for this course
SS3	I think the entrepreneurship and innovation course is one of the best learning experiences I've had
SS4R*	I think the content of the course needs to be improved and developed
SS5	I enjoyed studying this course
SS6	I am satisfied with the goals of the course
SS7	In general, I am satisfied with the educational/teaching content of the course
SS8	I am satisfied with the methods that are used in teaching the course
SS9	I am satisfied with the teaching conditions of the course
SS10	I am satisfied with the administrative conditions of the course
SS11	I am satisfied with the evaluation methods that are used in the course
Note(s): *R: Reverse question	

Roman (2014) to measure teaching methods. Five items were adapted from Gao *et al.* (2021) to measure teaching conditions. Three items were adapted from Gao *et al.* (2021) and Xiao *et al.* (2017) to measure teaching management. Five items were adapted from Rawlusyk (2018) to evaluate assessment methods. Finally, eleven items were adapted from prior studies (Ikram and Kenayathulla, 2023; Kanwar and Sanjeeva, 2022) to measure student satisfaction. Table 2

Table 2. Demographic profile of respondents

Category	Frequency	Percentage (100%)
<i>Gender</i>		
Male	195	40.1
Female	291	59.9
Total	486	100.0
<i>Academic year</i>		
First-year	50	10.3
Second year	234	48.1
Third year	168	34.6
Fourth-year	31	6.4
Fifth or above	3	0.6
Total	486	100.0
<i>Field of study</i>		
Humanities schools	70	14.4
Scientific schools	92	18.9
Health schools	324	66.7
Total	486	100.0
<i>Place of residence</i>		
Urban areas	445	91.6
Rural areas	41	8.4
Total	486	100.0

presents the items used to measure each construct in this study. Each question item was rated on a five-point Likert scale, with 1 representing full disagreement and 5 representing full agreement.

5. Data analysis

5.1 Statistical analysis

SPSS was chosen as the primary data analysis software for various reasons. First, the dataset includes structured survey data on a 5-point Likert scale, which is ideal for analysis using SPSS (Pallant, 2020). Second, this study focuses on descriptive statistics, reliability tests, regression analysis, and testing the direct effects—all of which SPSS handles effectively. Third, its availability and widespread use among researchers in the social and behavioral sciences, enhancing replicability (Pallant, 2020). Lastly, while PLS-SEM and other alternatives are options for modeling complex latent variable relationships, this study directly measures the variables and tests direct effects, rather than employing mediation of latent variables (Hair et al., 2021).

5.2 Common method bias

Harman's single-factor test was chosen to investigate common method bias (CMB), as it is more suitable for exploratory factor analysis (EFA) with SPSS than more complex procedures, which are more appropriate for structural equation modeling. The method is easy to understand and can reliably diagnose CMB, especially with self-reported cross-sectional data. Multiple factors emerged, with the first (highest) factor accounting for 28.5% of the total variance. Variance analyses indicated that no single factor accounted for more than 50% of the variance, and therefore CMB was not considered problematic (Podsakoff et al., 2003; Hair et al., 2021).

5.3 Validity and reliability

To ensure content validity, the questionnaire items were adapted from established measurement scales in prior studies. These constructs have demonstrated sufficient validity and reliability in the literature, supporting the appropriateness of the selected items. The questionnaire was translated into Arabic and reviewed by four subject matter experts to assess clarity, comprehensiveness, and relevance. Based on their feedback, necessary modifications were made, thereby confirming face validity.

To confirm construct validity, exploratory factor analysis (EFA) was conducted using all study items simultaneously. Principal component analysis was used as the extraction method, and varimax rotation was selected for the EFA. The Kaiser-Meyer-Olkin (KMO) test and Bartlett's test of sphericity were conducted before performing the EFA to assess sampling adequacy and factorability. The results of the KMO test demonstrated that the value obtained for each construct exceeded the cutoff of 0.50, indicating the appropriateness of the data for EFA. Additionally, the results of Bartlett's test of sphericity revealed that each construct had a statistically significant value ($p < 0.05$). The results of both tests confirmed the adequacy of the sample for conducting EFA (Sekaran and Bougie, 2016; Hair *et al.*, 2010).

We ensured that the items for each construct were loaded onto a single factor, with factor loadings exceeding 0.50 for each item and an eigenvalue greater than 1 for each construct, as presented in Table 3. Six items—TM2, TC2, EC6, TC1, EM5, and SS4—did not meet these criteria and were therefore removed.

Cronbach's alpha coefficient was used to evaluate the internal consistency and reliability of the study constructs. The findings revealed that the alpha values for the seven constructs exceeded 0.850, indicating excellent internal consistency and reliability of the measurement scales (Sekaran and Bougie, 2016). Table 4 presents Cronbach's alpha values, descriptive statistics, and the results of the KMO test and Bartlett's test of sphericity for all the constructs.

5.4 Hypothesis testing

Multicollinearity among the predictor variables was assessed to ensure that no issues could compromise the accuracy of the results (Hair *et al.*, 2010). Multicollinearity was evaluated using the variance inflation factor (VIF) and tolerance tests. The results indicated that VIF values ranged from 1.16 to 4.148, while tolerance values ranged from 0.241 to 0.356. These results confirm that all VIF values were below the recommended threshold of 5, and the tolerance values exceeded the acceptable minimum of 0.2 (Hair *et al.*, 2010).

To examine the impact of teaching quality dimensions on student satisfaction, multiple regression analysis was conducted. The results indicate that the proposed teaching quality dimensions account for approximately 81% of the variance in student satisfaction, as demonstrated by an R^2 value of 0.811 and an adjusted R^2 of 0.809. The model's overall significance is further supported by a high F-statistic (981.880, $p < 0.001$). However, the results indicate that course objectives ($\beta = 0.035$, $p = 0.349$) and teaching conditions ($\beta = -0.084$, $p = 0.050$) have no statistically significant impact on student satisfaction. Therefore, hypotheses H1 and H4 were not supported.

Furthermore, four of the six teaching quality dimensions examined had a statistically significant and positive impact on student satisfaction, as shown in Table 5. The strongest predictor was course content ($\beta = 0.293$, $p < 0.001$), highlighting the importance of providing relevant, engaging, and practical content in entrepreneurship and innovation education. Additionally, teaching management ($\beta = 0.254$, $p < 0.001$), evaluation techniques ($\beta = 0.250$, $p < 0.001$), and teaching methods ($\beta = 0.215$, $p < 0.001$) all had significant positive impacts on student satisfaction, supporting hypotheses H2, H3, H5, and H6.

5.5 Robustness checks

To ensure the stability and reliability of the regression results, the full sample was randomly divided into two equal subsamples ($n = 243$ each). Each subsample was independently

Table 3. EFA results for the study variables

Item	Course objective	Educational content	Teaching methods	Teaching conditions	Teaching management	Evaluation methods	Student satisfaction
CO1	0.702	—	—	—	—	—	—
CO2	0.547	—	—	—	—	—	—
CC1	—	0.911	—	—	—	—	—
CC2	—	0.716	—	—	—	—	—
CC3	—	0.900	—	—	—	—	—
CC4	—	0.765	—	—	—	—	—
CC5	—	0.505	—	—	—	—	—
CC6	—	deleted	—	—	—	—	—
CC7	—	0.610	—	—	—	—	—
TM1	—	—	0.569	—	—	—	—
TM2	—	—	deleted	—	—	—	—
TM3	—	—	0.676	—	—	—	—
TM4	—	—	0.766	—	—	—	—
TC1	—	—	—	deleted	—	—	—
TC2	—	—	—	deleted	—	—	—
TC3	—	—	—	0.705	—	—	—
TC4	—	—	—	0.783	—	—	—
TC5	—	—	—	0.768	—	—	—
TMA1	—	—	—	—	0.758	—	—
TMA2	—	—	—	—	0.587	—	—
TMA3	—	—	—	—	0.654	—	—
EM1	—	—	—	—	—	0.905	—
EM2	—	—	—	—	—	0.785	—
EM3	—	—	—	—	—	0.839	—
EM4	—	—	—	—	—	0.811	—
EM5	—	—	—	—	—	deleted	—
SS1	—	—	—	—	—	—	0.907
SS2	—	—	—	—	—	—	0.899
SS3	—	—	—	—	—	—	0.895
SS4	—	—	—	—	—	—	deleted
SS5	—	—	—	—	—	—	0.878
SS6	—	—	—	—	—	—	0.876
SS7	—	—	—	—	—	—	0.871
SS8	—	—	—	—	—	—	0.859
SS9	—	—	—	—	—	—	0.852
SS10	—	—	—	—	—	—	0.839
SS11	—	—	—	—	—	—	0.997
Eigenvalue	22.970	4.145	2.991	1.525	1.402	1.285	1.087
% of variance	21.318%	65.046%	68.502%	71.588%	74.021%	75.934%	77.803%

analyzed using the multiple regression model, allowing for a comparative assessment of results. The findings from both subsamples were highly consistent, showing consistent levels of statistical significance, regression coefficients, and R^2 values. This uniformity confirms the robustness, reliability, and reproducibility of the study's conclusions. Table 6 presents the results of the robustness check.

6. Discussion

The purpose of this study is to determine whether perceptions of teaching quality dimensions impact student satisfaction. The results indicate that dimensions of perception of teaching quality, including educational content, teaching methods, teaching management, and evaluation methods have a positive impact on student satisfaction.

Table 4. Descriptive statistics, reliability, and sampling adequacy for the study constructs

Construct	Mean	Std	Cronbach's alpha	KMO measure of sampling adequacy	Bartlett's test of sphericity (<i>p</i> -value)
Course objective	3.950	0.955	0.850	0.610	0.001
Course content	3.790	0.908	0.924	0.905	0.000
Teaching methods	3.850	0.652	0.913	0.710	0.000
Teaching conditions	3.853	1.00	0.884	0.695	0.000
Teaching management	3.688	1.03	0.913	0.748	0.000
Evaluation methods	3.610	1.01	0.906	0.822	0.000
Student satisfaction	3.612	0.974	0.950	0.940	0.000

Table 5. Results of regression analysis

Variable	β coefficient	<i>p</i> -value	Result
Constant	0.137	0.312	-
Course objective	0.035	0.349	Not supported
Course content	0.293	0.000	Supported
Teaching methods	0.215	0.000	Supported
Teaching conditions	-0.084	0.050	Not supported
Teaching management	0.254	0.000	Supported
Evaluation methods	0.250	0.000	Supported
<i>Model fit statistics</i>			
R	0.901	-	-
R^2	0.811	-	-
Adjusted R^2	0.809	-	-
F-statistic	981.880	0.000	-

Table 6. Robustness check: regression results for subsamples

Variable	Subsample 1 (<i>n</i> = 243)		Subsample 2 (<i>n</i> = 243)	
	β Coefficient	<i>p</i> -value	β Coefficient	<i>p</i> -value
Constant	-0.062	0.639	0.137	0.312
Course objective	0.013	0.803	0.054	0.318
Course content	0.260	0.000	0.303	0.000
Teaching methods	0.216	0.000	0.219	0.000
Teaching conditions	-0.085	0.050	-0.081	0.172
Teaching management	0.286	0.000	0.245	0.000
Evaluation methods	0.289	0.000	0.239	0.000
<i>Model fit statistics</i>				
R	0.910	-	0.891	-
R^2	0.829	-	0.793	-
Adjusted R^2	0.824	-	0.788	-
F-statistic	189.510	0.000	151.319	0.000

These findings partially align with previous research (Gao *et al.*, 2021; Kim and Park, 2023). However, no significant impact was found for course objectives and teaching conditions on student satisfaction.

Our findings reveal a positive impact of the educational content on student satisfaction. This result is consistent with the outcomes of some prior studies (Ikram *et al.*, 2025; Darawong and Widayati, 2022). This finding is unsurprising, as when the content is relevant, clear, organized, and logically structured, it increases students' understanding and engagement, leading to satisfaction with the provided content and materials (Gao *et al.*, 2021; Roman, 2014). The strong positive relationship between the course content and student satisfaction underscores the need to provide relevant, organized, and comprehensive materials. Students are more satisfied when the course content is targeted to their learning goals and adds value to their understanding of the subject matter. The results indicate that students prefer course content that is relevant to real-world situations. For example, entrepreneurship and innovation courses with practical contexts, such as case studies, business simulations, and industry-relevant examples, help students prepare for their future professions.

Our findings confirm that teaching methods positively affect student satisfaction, aligning with earlier studies (Akram *et al.*, 2022; Pandita and Kiran, 2023). The positive relationship between teaching methods and student satisfaction suggests that varied, interesting methods of instruction contribute to a more active and rewarding educational experience. Methods that are participatory, such as group work, hands-on projects, and problem-based learning, can increase student satisfaction in meaningful ways. This implies that an active learning approach makes learning more engaging, while methods that accommodate different learning styles enhance accessibility. In contrast to Darawong and Widayati (2022) and Ikram and Kenayathulla (2023), who also found that teaching methods positively influenced student satisfaction, our study was conducted specifically in the context of entrepreneurship and innovation courses, making our findings particularly relevant to this field.

The study results show that teaching management positively impacts student satisfaction, consistent with previous studies (Darling-Hammond, 2021; Darling-Hammond *et al.*, 2019). The strong positive impact of teaching management on student satisfaction indicates the importance of well-managed classes. Good time management, communicative clarity, and lesson structure enhance the classroom experience and facilitate learning, which ultimately leads to increased student satisfaction (Darling-Hammond, 2021; Darling-Hammond *et al.*, 2019). Effective course organization, resource allocation, and learning environments contribute to students' overall satisfaction. Our study differs from past research, which has emphasized various pedagogical aspects rather than organizational factors.

Similarly, evaluation methods positively affect student satisfaction, aligning with previous literature (Mate and Weidenhofer, 2022; Rigopoulos, 2022). The ways of assessing student learning (including exams, projects, and continuous assessment) impact how the student engages with the course and the extent to which students learn. Fair and transparent assessment processes can positively impact student performance. This indicates that when evaluation systems are well-adapted and aligned with evaluation techniques and student expectations, they enhance satisfaction. Additionally, transparency and consistency between evaluation methods and learning objectives increase students' trust in their abilities to learn. Unlike prior studies that examined evaluation methods in general contexts (Mate and Weidenhofer, 2022; Rigopoulos, 2022), our research focuses on entrepreneurship and innovation, providing a specialized perspective.

On the other hand, course objectives did not significantly impact student satisfaction, contradicting previous research (Darawong and Widayati, 2022; Harsasi and Sutawijaya, 2018). While the course objectives are critical for developing the framework of the course, the findings imply that they have little impact on student satisfaction. This may be due to students engaging more with what they actually do within the course rather than with the formal, stated course objectives. Three possible explanations may account for this. First, in many educational contexts, greater emphasis is placed on measurable outcomes (e.g. grades, degrees) than on the

learning process itself, leading students to value tangible results over stated objectives. Second, in some Middle Eastern institutions, where students are expected to follow instructions without question, they may prioritize completing tasks and respecting instructors over engaging with course goals. Finally, limitations in the measurement tools, such as poorly worded items or cultural preferences for practical teaching over theoretical goals, may have influenced the findings.

Another key finding was the insignificant and negative impact of teaching conditions on student satisfaction. This result contrasts with prior literature (Alqasa and Afaneh, 2022; Gao *et al.*, 2021), which found a significant positive effect. Three possible explanations exist. First, classroom environments may lack proper air conditioning and seating arrangements, making conditions uncomfortable. Second, broader institutional or structural challenges within the higher education sector in Jordan, such as insufficient infrastructure, limited access to technology, and overcrowded classrooms, may hinder the overall quality of the learning environment, thereby diminishing student satisfaction even in the presence of effective teaching. Third, the absence of dedicated spaces for innovation, including manufacturing labs, innovation hubs, or think tank facilities, restricts students' opportunities to develop and test their ideas. Although teaching conditions are expected to impact student satisfaction, the negative effect observed here highlights the importance of ensuring that the environment facilitates the teaching process. However, its impact may not be as strong as other factors such as subject content and teaching methods.

7. Conclusions, implications, and limitations

7.1 Conclusions

This study examines the influence of the teaching quality dimensions on student satisfaction in the Fundamentals of Entrepreneurship course at the University of Jordan. It uses a conceptual model called the Jordanian Student Satisfaction Model and the Input-Output Model to analyze and test hypotheses related to teaching quality and its impact on student satisfaction. The results reveal that teaching content, methods, management, and evaluation methods are the primary dimensions positively affecting student satisfaction. However, teaching conditions were found to have a significant negative effect on satisfaction, while course objectives showed no significant impact. The study highlights the need for universities to improve course objectives and focus on enhancing classroom conditions and logistics to boost student satisfaction.

7.2 Theoretical and social implications

This research offers important theoretical contributions to the literature on teaching quality and student satisfaction in higher education. By introducing two new factors—course objectives and evaluation methods—as key elements in the entrepreneurship and innovation education context. Investigating the study within the context of the Jordanian higher education system fills multiple gaps in the literature, which is based mainly on Western contexts. In addition, the study brings an interdisciplinary view by drawing on concepts from educational quality and entrepreneurship education and therefore contributes to creating a domain-specific model of student satisfaction that is culturally relevant. Finally, the study provides a new model for the Jordan setting, namely the Jordanian Student Satisfaction Model for Higher Education (JSSMHE) which enhances the existing literature in the field.

7.3 Managerial implications

Regarding practical implications, universities should consistently update and adapt their curriculum to align with industry requirements and enhance teaching quality. Academic institutions are required to develop curricula that are both responsive to the needs of students and in line with the realities of modern entrepreneurship. Stronger integration of case studies,

real-world examples, and project-based learning that takes market dynamics into account can help achieve this. Formal collaborations with industry sectors can facilitate guest speakers, internships, and joint workshops, as well as provide access to valuable resources, all of which improve the relevance and application of learning. To improve classroom practice, universities should provide faculty with support, training, collaboration opportunities, peer review, and professional development sessions. Addressing infrastructural and environmental barriers that could impede student engagement becomes crucial. Whether through international development grants, private sector assistance, or public funding, universities and legislators should investigate sustainable funding methods to enhance learning environments. Additionally, physical and environmental considerations must be addressed to create a more conducive learning atmosphere. Universities should refine evaluation methods to ensure they accurately reflect student performance and provide constructive feedback. Decision-makers in educational institutions should prioritize the professional development of lecturers, enabling them to effectively integrate curriculum objectives and implement engaging teaching methods that foster a positive learning experience. Well-trained instructors who can translate educational goals into dynamic activities are more likely to improve student outcomes and enhance satisfaction. Furthermore, institutions should invest in the development of educational infrastructure, such as smart classrooms and efficient air conditioning and heating systems, as these factors significantly impact the satisfaction of both students and lecturers.

Societally, this study's enhancement of student satisfaction in higher education—especially in areas like entrepreneurship—can support national objectives related to economic growth, employment, and innovation on a broader social scale. Satisfied students are more likely to stay involved, cultivate entrepreneurial mindsets, and enter the workforce with the abilities and self-assurance required to start or support businesses. As a result, the study has implications for Jordan and similar countries' public policy planning, national development, and educational practice.

7.4 Limitations and future studies

The study includes several limitations that should be acknowledged. First, the measurement of the “course objectives” construct, which was evaluated with just two items, is one significant limitation. Even though these items demonstrated satisfactory psychometric attributes, the scale and internal consistency of the construct may not be fully captured by using fewer than three items. Future studies should focus on constructing and validating more robust multi-item scales for this construct. Second, the adoption of the 0.50 cutoff point in EFA—although considered acceptable—aims to strike a balance between maintaining conceptual consistency and structural correctness. Hence, to increase accuracy and dependability, future studies should adopt a more cautious cut-off point. Third, the study's only emphasis on the Jordanian context limits the findings' generalizability due to cultural constraints. Future research should widen its reach by conducting empirical studies in a variety of cultural settings. Fourth, the study focused on a single university in Jordan. Future studies should include multiple universities in Jordan to provide a comprehensive view. Finally, while our study employed well-established and validated measurement scales to ensure reliability and facilitate comparison with international research, this reliance on pre-existing literature may limit the contextual specificity of the findings. Future research should develop and validate measurement items specifically tailored to the Jordanian higher education context, which would further enhance the originality and cultural relevance of the Jordanian Student Satisfaction Model for Higher Education (JSSMHE).

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